

macro strategy (e.g., $POV = 5\%$) utilizing a larger number of limit orders to avoid crossing the bid-ask spread, since there is ample time for these limit orders to be lifted in the market.

In most situations it will be appropriate to use a combination of limit, market, floats and reserve orders. For example, suppose the specified macro-level optimal strategy is a POV rate of 15%. Here a micro-level algorithm may submit limit orders to the market for execution better than the mid-quote for as long as the actual POV rate is consistent with 15% of market volume, but once the algorithm starts lagging behind the specified rate or some specified tolerance level it would submit appropriately sized and spaced market orders to be more aggressive and adhere to the 15% rate. A reserve order could also be used to automatically replenish limit orders at favorable prices. Some of the more advanced micro pricing strategies utilize real-time data, prices and quotes, order book, and recent trading activity to forecast very short-term price trends and provide probabilistic estimates surrounding the likelihood that a limit order will execute within a certain period of time.

Limit Order Models

Limit order models determine the appropriate mix of limit and market orders to best adhere to the higher level macro goals. The limit order model is a probabilistic model that takes into account current market conditions, price momentum, order book information, macro goal and timing. Traditionally, limit order models will determine the probability that an order will execute in the market at a stated price and within a stated amount of time. The limit order model here is a modified limit order model with the output being a mix of prices and share quantities to ensure completion by the end of the time period (or at least a high enough likelihood of completion) rather than a probability estimate of executing at a specified price point.

For example, if the optimal trading rate specified in the macro-level decision for a buy order is $POV = 10\%$ and we forecast 10,000 shares will be traded in the next 1 minute, then we will need to execute 1000 shares in the next 1 minute to adhere to our 10% POV rate. If the current market is \$30.00–\$30.10 the limit order model may determine that the most cost effective mix of prices, order type, and share quantity to trade these 1000 shares in the next 1 minute is:

- Limit order, 200 shares at \$29.95
- Limit order, 300 shares at \$30.00
- Limit order, 300 shares at \$30.05
- Market order, 200 shares at \$30.10